

S M MAZHARUL HOQUE CHOWDHURY

PhD Student, Machine Learning & Accessibility Researcher
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PROFESSIONAL SUMMARY

Highly qualified PhD Student specializing in Accessibility Research, Machine Learning (ML), Natural Language Processing (NLP), and Human-Computer Interaction (HCI). Seeking a Machine Learning/AI Internship or job to contribute to projects involving large language models, adaptive systems, and accessibility. Expertise in the full ML lifecycle, from advanced deep learning and prototyping interactive systems to conducting rigorous academic research. Proficient in Python, PyTorch, TensorFlow, and statistical modeling.

HARD TECHNICAL SKILLS

Programming	Python, C, C++
ML/DL Frameworks	PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy
ML Domains	NLP (LLMs/RNNs), Accessibility, Adaptive Systems, Multimodal Sensing, Fairness & Bias Detection
Research Experience	Accessibility, HCI, Artificial Intelligence, NLP, Machine Learning, Computer Vision, IoT.

SOFT SKILLS & LEADERSHIP

- Technical Communication: Proven ability to convey complex ML concepts to diverse technical and non-technical audiences (via teaching and international publications).
- Ability to Collaborate & Mentorship: Experienced in cross-disciplinary teamwork (research projects) and leading/mentoring undergraduate students (GTA role).
- Problem Solving Skill: Strong capability in critical analysis, debugging ML pipelines, and prioritizing ethical considerations (Fairness, Privacy) in system design.
- Decision making: Demonstrated capability in evaluating complex trade-offs to guide project direction and achieve optimal ethical and technical outcomes.
- Research Tenacity & Ambition: Highly committed and ambitious researcher, utilizing systematic investigations and technical rigor to address complex, open-ended problems and drive projects beyond expected academic requirements.

EDUCATION

PhD in Computer Science and Engineering | University of North Texas, Denton, TX (2023–Present)

- Research: Adaptive, gamified ML systems for accessibility (Advisor: Dr. Stephanie Ludi). Focus on Personalization, Fairness, and Behavioral Modeling.

M.Sc. in Computer Science | Jahangirnagar University (2018–2019)

MACHINE LEARNING & ACCESSIBILITY PROJECTS

Bengali LLM Prototyping: Bi-directional RNN Text Generation

- Developed a foundational deep learning model using Bi-directional Recurrent Neural Networks (RNNs) to generate complex Bengali text, demonstrating core proficiency in sequence modeling relevant to LLMs.

ML for Fairness: Hate Speech Detection in Low-Resource Language

- Implemented a machine learning classification pipeline to detect and categorize hate speech, addressing core research challenges in fairness and content moderation.

PEER-REVIEWED PUBLICATIONS

- Bengali text generation using bi-directional RNN (2019) *Research Focus: Sequence Modeling and Language Generation.*
- Implementation of machine learning to detect hate speech in Bangla language (2019) *Research Focus: Ethical AI and Fairness in ML applied to content moderation.*
- Gesture-based smart wheelchair for assisting physically challenged people (2021) *Research Focus: Accessibility Research and Interactive Systems design.*
- Wikipedia Empowered Natural Language Interface for Web Search (2024) *Research Focus: Advanced NLP for Information Extraction.*

WORK EXPERIENCE

Graduate Teaching Assistant (GTA) | University of North Texas (Aug 2023 – Present)

- Mentored and provided technical guidance into undergraduate students in core CS and object-oriented programming lab environments.

Faculty (O/A Level Computer Science) | Abdul Kadir Molla International School (Mar 2021 – Jun 2023)

- Instructed students in advanced Computer Science curricula, achieving high pass rates in O/A Level examinations.

Research Associate | Apurba-DIU Research & Development Lab (Jan 2021 – Mar 2021)

- Contributed to R&D projects, gaining practical experience in integrating research prototypes into functional systems.

Lecturer | Daffodil International University (Sep 2018 – Apr 2020)

- Delivered comprehensive lectures and managed practical lab sessions for undergraduate CSE students.

RESEARCH AREAS & FOCUS

- Adaptive Systems: Personalized ML models and decision-making theory for dynamic, user-centered applications.
- Ethical AI: Focusing on fairness, bias mitigation, and privacy-preserving ML techniques.
- Multimodal Fusion: Integrating data from vision, text, and sensors to inform ML models for practical outcomes.